



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

Data Communication

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OEL Report

Binary Vector Generation and Digital Data Transmission using BASK Modulation with FDM

Supervised By

MOHAMMAD ASADUZZAMAN KHAN

Submitted By :

Name	ID
1. MD Shahriar Laskher	23-51207-1
2. Shahariar Khan	23-51222-1
3. Md. Saikot Hossain	23-51242-1
4. T.A Nahian	23-51257-1
5. Md.Mosharof Hossain Khan	23-51259-1

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Title:

Binary Vector Generation and Digital Data Transmission using BASK Modulation with FDM.

Objectives:

The objective of this experiment is to understand and demonstrate the complete process of digital data transmission using ASCII encoding, Binary Amplitude Shift Keying (BASK) modulation, and Frequency Division Multiplexing (FDM). Specifically, the goals are:

- a) To convert surnames of group members into binary vectors using ASCII codes.
- b) To represent the binary data in time domain using unipolar NRZ signaling.
- c) To apply BASK modulation with different carrier frequencies for each binary vector.
- d) To combine the modulated signals into a single composite signal using FDM and introduce Gaussian noise.
- e) To perform demultiplexing and demodulation in order to recover the original binary signals.
- f) To decode the recovered binary data back into ASCII characters and verify the accuracy of transmission.

Theory:

A. Frequency Division Multiplexing (FDM):

Frequency Division Multiplexing (FDM) is a technique used to transmit multiple signals simultaneously over a single communication channel by allocating each signal a distinct frequency band. In FDM, each signal is modulated onto a separate carrier frequency, ensuring that the frequency spectra of the signals do not overlap. At the receiver, bandpass filters are used to separate the signals, and demodulation is performed to recover the original data. FDM is widely used in radio, television, and telecommunication systems where multiple signals must be transmitted over a shared medium.

B. Binary Amplitude Shift Keying (BASK) Modulation:

Binary Amplitude Shift Keying (BASK) is a digital modulation technique in which the amplitude of a carrier signal is varied according to the binary data. For a binary '1', the carrier signal is transmitted with a fixed amplitude, while for a binary '0', the carrier is either turned off or transmitted with zero amplitude. BASK is simple to implement and suitable for transmitting digital data over a continuous-time analog channel. Its performance is affected by noise, as amplitude variations can be distorted during transmission.

C. BASK Demodulation

BASK demodulation is the process of recovering the original binary data from a BASK-modulated signal. This is typically achieved using coherent or envelope detection techniques. In coherent detection, the received signal is multiplied by a synchronized carrier and passed through a low-pass filter to recover the baseband data. In envelope detection, the amplitude of the received signal

is detected and compared against a threshold to determine the transmitted binary values. Proper demodulation requires accurate carrier synchronization and sufficient signal-to-noise ratio (SNR) to minimize errors.

Result:

```
function dn = asc2bn(txt)
dec=double(txt)
p2=2.^(0:-1:-7) % 2^0,2^-1,.....,2^-7
B=mod(floor(p2'*dec),2)
dn=reshape(B,1,numel(B));
end
function txt = bin2asc(dn)
L=length(dn);
L8=8*floor(L/8);
B=reshape(dn(1:L8),8,L8/8);
p2=2.^(0:7); %power of 2
dec=p2*B;
txt=char(dec);
end
clc;
clear all;
close all;
% Task 1
surname1='KHAN'
surname2='NAHIAN'
x1=asc2bn(surname1);
x2=asc2bn(surname2);
disp(' Binary information at Transmitter (x1):');
disp(x1);
disp(' Binary information at Transmitter (x2):');
disp(x2);
%Task 2: Digital NRZ signals (like given style)
bp = 1e-3;          % bit period
% For x1
bit1 = [];
for n = 1:length(x1)
    if x1(n) == 1
        se = 5 * ones(1,100); % amplitude 5V for bit=1
    else
        se = zeros(1,100);    % amplitude 0V for bit=0
    end
    bit1 = [bit1 se];         % concatenate
end
```

```
t1 = bp/100 : bp/100 : 100*length(x1)*(bp/100);
```

```
figure(1);  
subplot(2,2,1);  
plot(t1,bit1,'b','LineWidth',2.5); grid on;  
axis([0 bp*length(x1) -0.5 6]);  
ylabel('Amplitude (V)'); xlabel('Time (sec)');  
title(['Unipolar NRZ signal for ', surname1]);
```

```
%For X2
```

```
bit2 = [];  
for n = 1:length(x2)  
    if x2(n) == 1  
        se = 5 * ones(1,100);  
    else  
        se = zeros(1,100);  
    end  
    bit2 = [bit2 se];  
end  
t2 = bp/100 : bp/100 : 100*length(x2)*(bp/100);  
subplot(2,2,2);  
plot(t2,bit2,'r','LineWidth',2.5); grid on;  
axis([0 bp*length(x2) -0.5 6]);  
ylabel('Amplitude (V)'); xlabel('Time (sec)');  
title(['Unipolar NRZ signal for ', surname2]);
```

```
% Task 3: BASK Modulation with f1=2 Hz, f2=4 Hz
```

```
A1 = 5;  
A2 = 0;  
br = 1/bp;  
f1 = 2000;  
t2 = bp/99 : bp/99 : bp;  
ss = length(t2);  
m1 = [];  
for i = 1:1:length(x1)  
    if (x1(i) == 1)  
        y = A1 * cos(2*pi*f1*t2);
```

```

else
    y = A2 * cos(2*pi*f1*t2);
end
m1 = [m1 y];
end
t3_1 = bp/99 : bp/99 : bp*length(x1);
subplot(2,2,3);
plot(t3_1, m1, 'b','LineWidth',1.5);
axis([0 bp*length(x1) -6 6]);
xlabel('time(sec)'); ylabel('amplitude(volt)');
title(['BASK Modulated Signal (x1) - ', surname1, ' (f1=2 Hz)']);

f2 = 4000;
m2 = [];
for i = 1:1:length(x2)
    if (x2(i) == 1)
        y = A1 * cos(2*pi*f2*t2);
    else
        y = A2 * cos(2*pi*f2*t2);
    end
    m2 = [m2 y];
end
t3_2 = bp/99 : bp/99 : bp*length(x2);

subplot(2,2,4);
plot(t3_2, m2, 'r','LineWidth',1.5);
axis([0 bp*length(x2) -6 6]);
xlabel('time(sec)'); ylabel('amplitude(volt)');
title(['BASK Modulated Signal (x2) - ', surname2, ' (f2=4 Hz)']);

% Task 4: Composite (FDM) + Noise
if length(m1) > length(m2)
    m2(length(m1)) = 0; % Zero-padding shorter signal
elseif length(m2) > length(m1)
    m1(length(m2)) = 0; % Zero-padding shorter signal
end

```

```

composite_signal = m1 + m2;          % FDM composite
t_comp = bp/99 : bp/99 : bp/99*length(composite_signal);

disp(' Message transmitted through Transmission medium');
SNR_dB = 10;                          % desired SNR
sig_pwr = mean(composite_signal.^2); % signal power
noise_pwr = sig_pwr / (10^(SNR_dB/10));
noise = sqrt(noise_pwr) * randn(size(composite_signal));
Rec = composite_signal + noise;

figure(2);
subplot(3,1,1);
plot(t_comp, Rec,'c','LineWidth',1.2);
axis([0 bp*length(x1) -6 6]);
xlabel('time(sec)');
ylabel('amplitude(volt)');
title('Received Composite Signal with Noise');
grid on;

```

% Task 5: Demodulation

```

mn1 = []; % recovered bits for user1
mn2 = []; % recovered bits for user2
ref1 = cos(2*pi*f1*t2);
ref2 = cos(2*pi*f2*t2);
for n = ss:ss:length(Rec)
    seg = Rec(n-(ss-1):n); % take one bit segment
    % multiply with reference carriers
    z1 = trapz(t2, seg .* ref1);
    z2 = trapz(t2, seg .* ref2);
    % scale
    zz1 = round((2*z1)/bp);
    zz2 = round((2*z2)/bp);
    % threshold detection (A1=5, A2=0 so mid = 2.5)
    if zz1 > 2.5
        a = 1;
    else
        a = 0;
    end
    mn1 = [mn1 a];

```

```

    if zz2 > 2.5
        b = 1;
    else
        b = 0;
    end
    mn2 = [mn2 b];
end
disp(' Binary information at Receiver (x1):');
disp(mn1);
disp(' Binary information at Receiver (x2):');
disp(mn2);

bit1 = [];
for n=1:length(mn1)
    if mn1(n)==1
        se=5*ones(1,100);
    else
        se=zeros(1,100);
    end
    bit1=[bit1 se];
end
t_rec1 = bp/100:bp/100:100*length(mn1)*(bp/100);

bit2 = [];
for n=1:length(mn2)
    if mn2(n)==1
        se=5*ones(1,100);
    else
        se=zeros(1,100);
    end
    bit2=[bit2 se];
end
t_rec2 = bp/100:bp/100:100*length(mn2)*(bp/100);
subplot(3,1,2);
plot(t_rec1, bit1,'b','LineWidth',2.5);
axis([0 bp*length(mn1) -0.5 6]);
xlabel('time(sec)'); ylabel('amplitude(volt)');

```

```

title('Recovered NRZ Signal (x1)');
grid on;

subplot(3,1,3);
plot(t_rec2, bit2,'r','LineWidth',2.5);
axis([0 bp*length(mn2) -0.5 6]);
xlabel('time(sec)'); ylabel('amplitude(volt)');
title('Recovered NRZ Signal (x2)');
grid on;

```

%Task:6

```

Received_Message1 = bin2asc(mn1);
Received_Message2 = bin2asc(mn2);

```

```

disp('Recovered message from x1:');
disp(Received_Message1);

```

```

disp('Recovered message from x2:');
disp(Received_Message2);

```

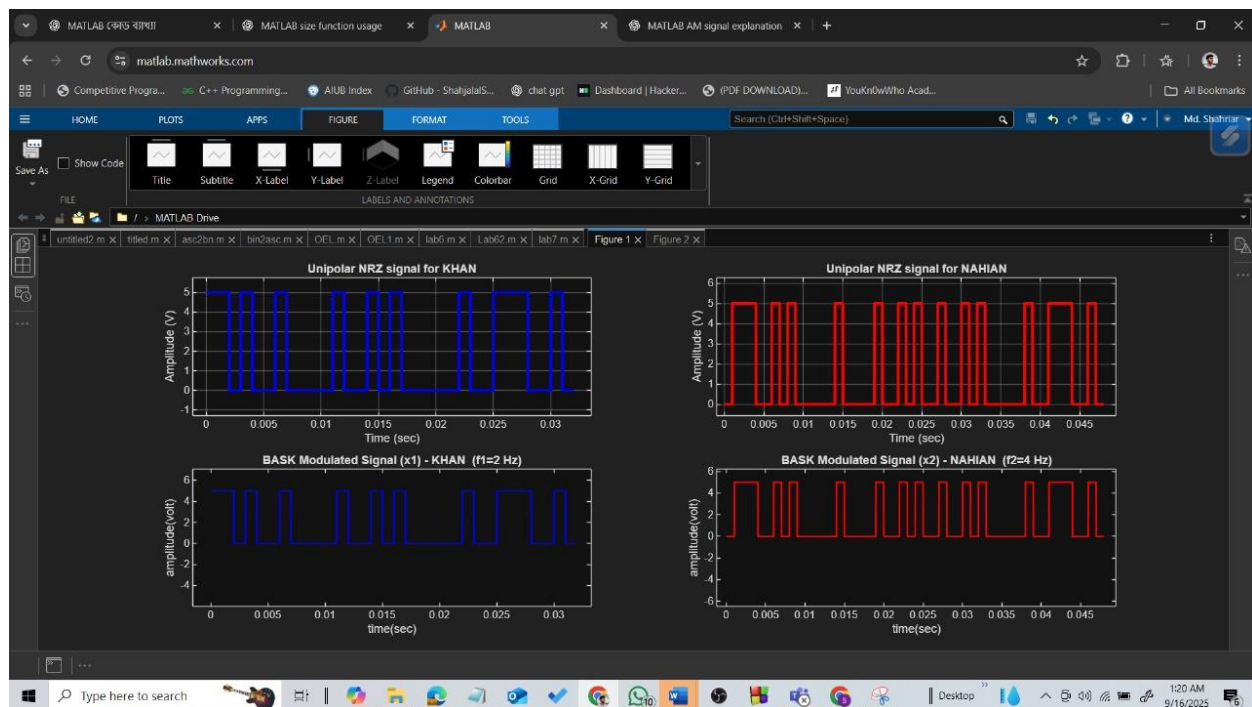


Figure 1: Figure for 2 and 4 Hz frequency

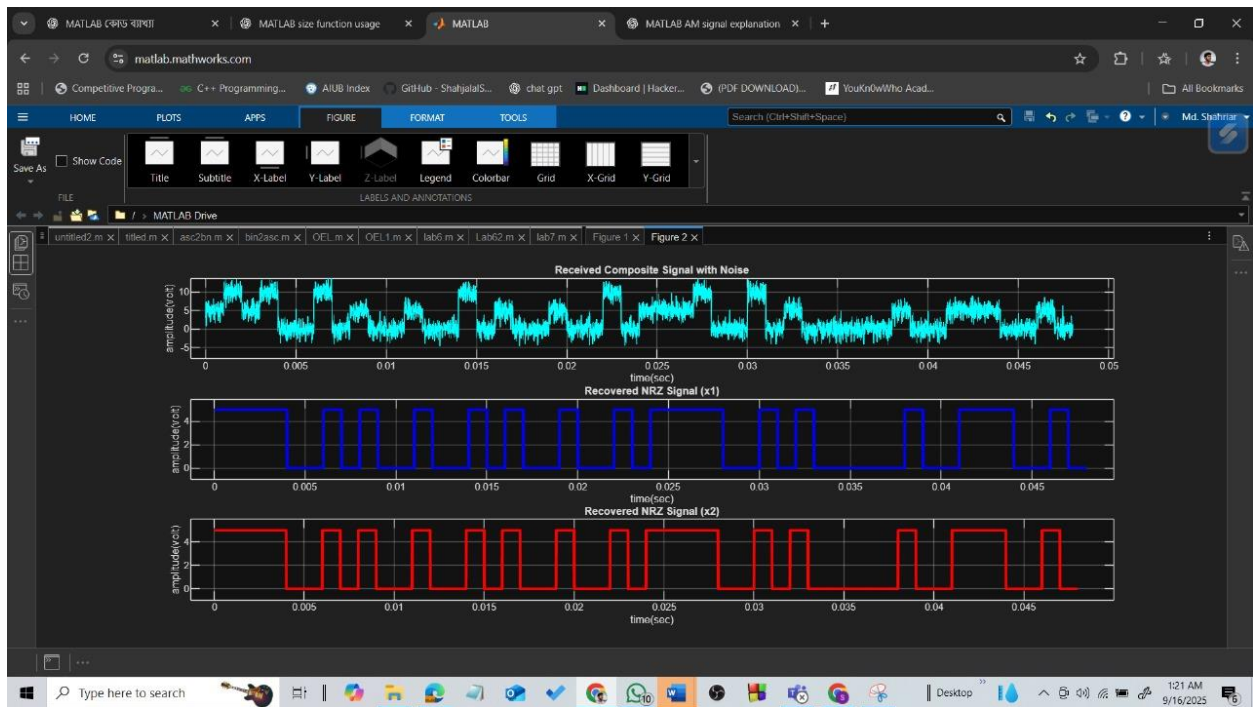


Figure 02: Figure for 2 and 4 Hz frequency

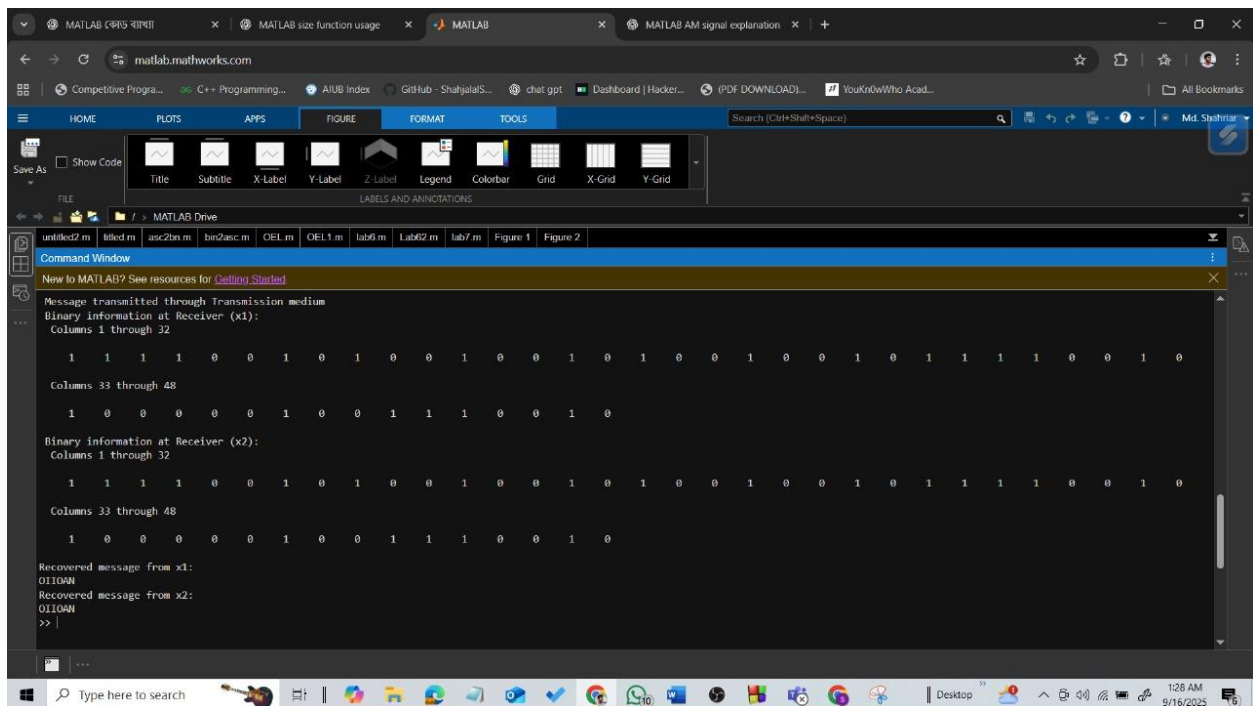


Figure 03: Figure for 2 and 4 Hz frequency

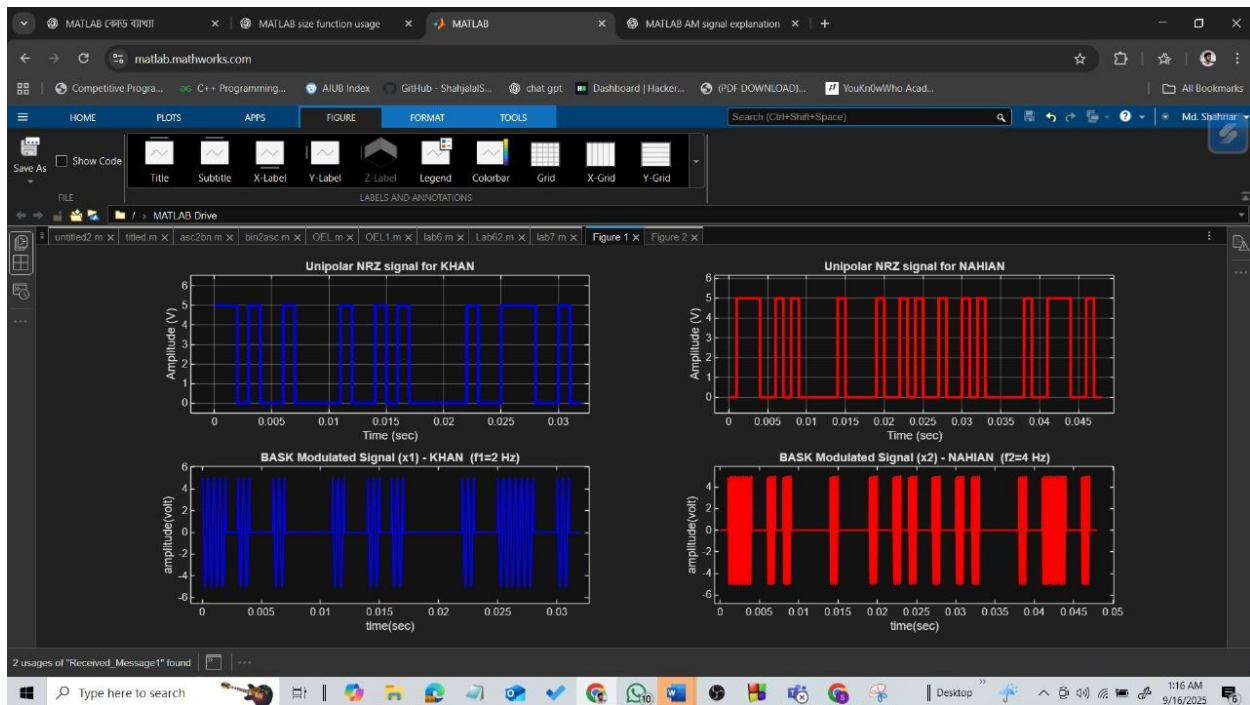


Figure 04: Figure for 2000 and 4000 Hz frequency

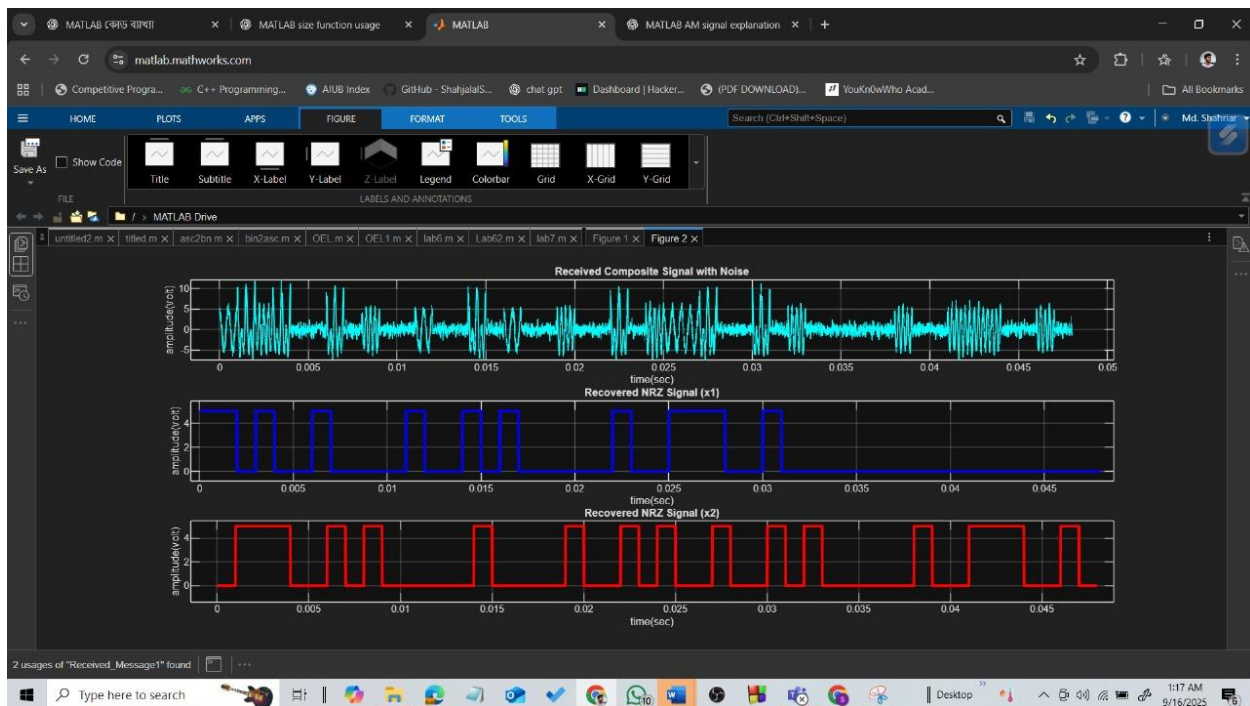


Figure 05: Figure for 2000 and 4000 Hz frequency

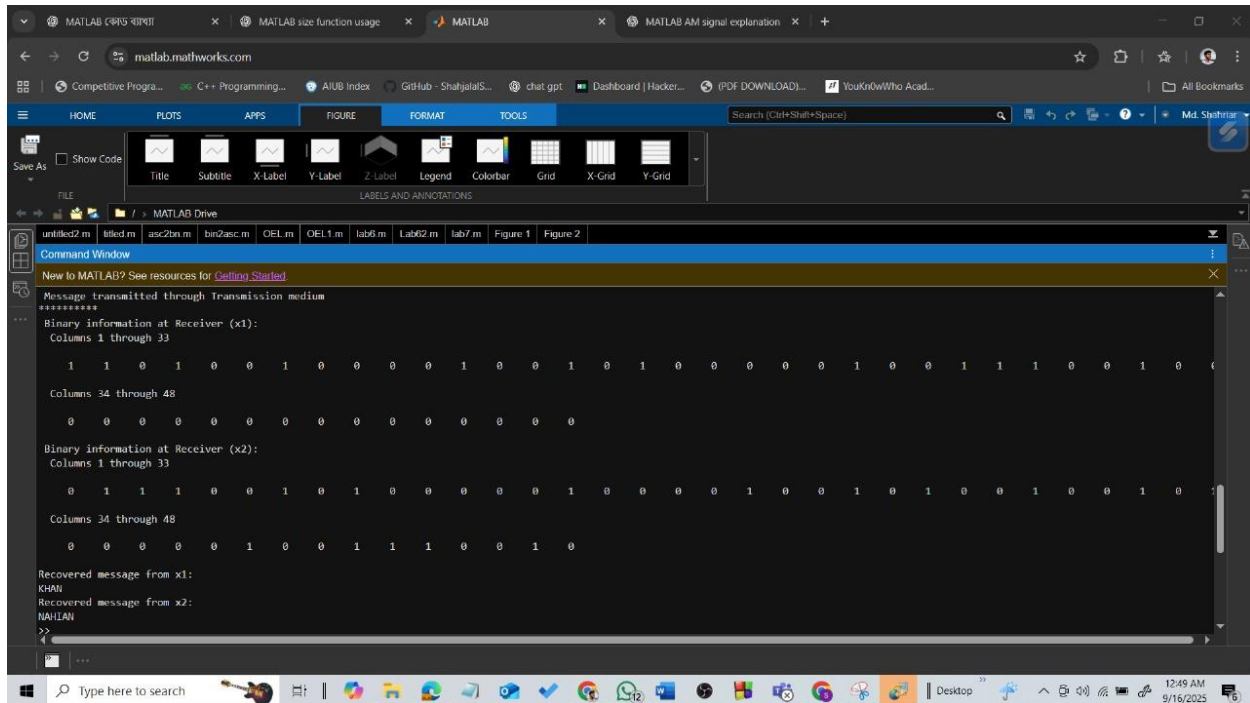


Figure 06: Figure for 2000 and 4000 Hz frequency

The experiment generated several MATLAB figures illustrating different stages of digital data transmission. In **Figure 1**, the original binary vectors $x1$ and $x2$ were plotted using unipolar NRZ signaling with a bit duration of 1 ms. The waveforms accurately represented the binary sequences corresponding to the ASCII-encoded surnames. Following this, BASK modulation was applied, where $x1$ was modulated with a 2 Hz carrier and $x2$ with a 4 Hz carrier. The modulated signals were clearly visualized in subplots, demonstrating the variation of the carrier amplitude according to the binary input data.

In **Figure 2**, the two BASK-modulated signals were combined using Frequency Division Multiplexing (FDM) to form a composite signal. Gaussian noise was added to simulate real-world transmission conditions. Despite the noise, the demultiplexing process effectively separated the individual BASK-modulated signals. After BASK demodulation, the original binary sequences $x1$ and $x2$ were successfully recovered and plotted in unipolar NRZ format. The recovered signals matched the transmitted binary data, confirming the accuracy of the transmission process.

The effect of noise and Signal-to-Noise Ratio (SNR) was observed during demodulation. At high SNR levels, the demodulated signals closely matched the original binary vectors, and no errors were detected. As the noise level increased or SNR decreased, small amplitude variations appeared in the demodulated signals, which could lead to bit errors if the SNR fell below a critical threshold. This highlighted the importance of maintaining a sufficient SNR for reliable BASK demodulation and accurate signal recovery. Proper selection of carrier frequencies and filtering during FDM also contributed to minimizing interference between channels.

Result Analysis:

The experiment demonstrated the complete process of digital data transmission using Binary Amplitude Shift Keying (BASK) modulation combined with Frequency Division Multiplexing (FDM) in MATLAB. Initially, surnames of two group members were converted into binary vectors using the ASCII table, generating the digital signals x1 and x2 in synchronous serial transmission format. These vectors were represented in the time domain using unipolar NRZ signaling with a bit duration of 1 ms, providing a clear visualization of the digital input signals. BASK modulation was applied to each signal. Initially, x1 was modulated with a carrier frequency of 2 Hz and x2 with 4 Hz. However, it was observed that at such low frequencies, the modulated signals were not clearly distinguishable in the plots. Therefore, higher carrier frequencies of 2000 Hz and 4000 Hz were used for x1 and x2, respectively, resulting in clearly visible BASK-modulated signals. The amplitude of the carrier varied according to the binary input, and the modulated signals were successfully displayed using subplots, illustrating the modulation process effectively. The two BASK-modulated signals were then combined into a composite signal using FDM, and Gaussian noise was added to simulate real-world transmission conditions. The time-domain representation of the composite signal confirmed the proper multiplexing of the two data streams. At the receiver end, the composite signal was demultiplexed into its constituent BASK-modulated signals. Demodulation of each signal recovered the original digital sequences x1 and x2. These signals were plotted in unipolar NRZ format, and the recovered binary vectors were decoded back into ASCII characters. The decoded output matched the original surnames, verifying the accuracy of the transmission process. The MATLAB simulation provided a comprehensive understanding of how digital data can be transmitted efficiently using BASK modulation with FDM. It highlighted the effects of noise, the importance of carrier frequency selection, and the ability of BASK modulation to preserve signal integrity in a multiplexed system. Using higher carrier frequencies significantly improved the clarity of signal visualization, making the modulation and multiplexing processes easier to analyze.

Conclusion:

In this experiment, digital data transmission using BASK modulation with FDM was successfully demonstrated in MATLAB. Binary vectors representing two surnames were generated, modulated, multiplexed, and transmitted over a composite channel with added noise. The signals were accurately demultiplexed, demodulated, and decoded back to their original characters, confirming the effectiveness of BASK and FDM in reliable digital communication. The study highlighted the importance of carrier frequency selection and noise handling for successful signal recovery.

References:

- [1] Lab Manual
- [2] MATLAB